

Windows Server + Active Directory Home Lab

Lab portfolio document for IT Help Desk practice

Name	Marc Moran
Lab Focus	Windows Server, Active Directory, DNS, Windows 11 domain client, and help desk account tasks
Domain	marclab.local
Server	Server01 - Domain Controller and DNS
Client	Client01 - Windows 11 domain-joined workstation
Virtualization	Hyper-V

Purpose

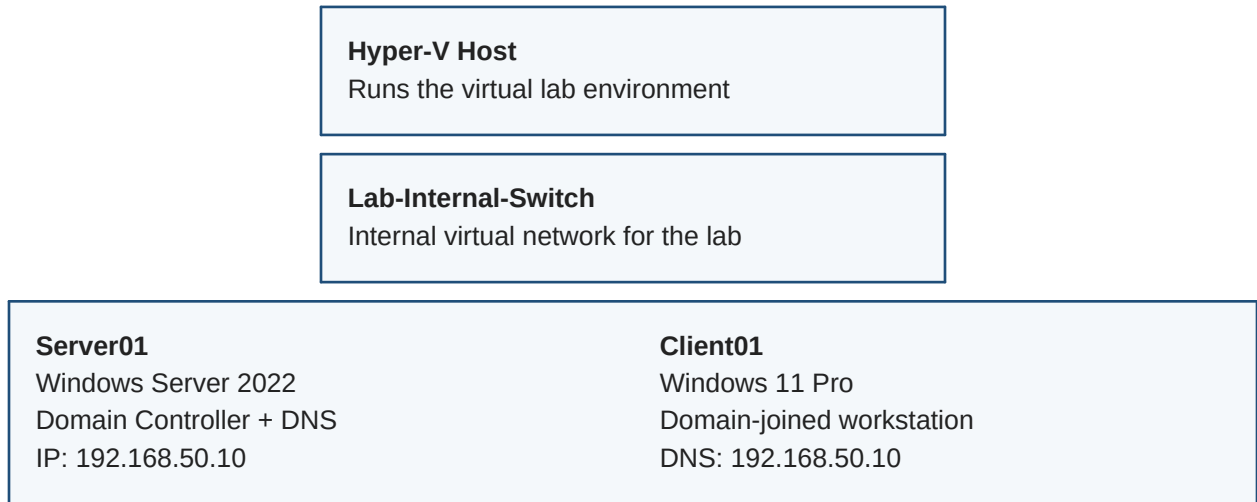
This lab was built to gain practical IT Help Desk experience with common Windows domain administration tasks. The environment simulates a small business network using a Windows Server domain controller and a Windows 11 client joined to the domain.

Skills Demonstrated

- Built a Windows Server domain lab in Hyper-V.
- Configured a domain controller for marclab.local.
- Joined a Windows 11 client to the domain.
- Verified network connectivity, DNS resolution, and domain login.
- Created and organized users, computers, groups, and OUs.
- Practiced password resets, account unlocks, account disablement, and new user onboarding.
- Documented help desk-style tickets and troubleshooting steps.

Lab Topology

The lab uses an internal Hyper-V network so the virtual machines can communicate with each other in an isolated environment.



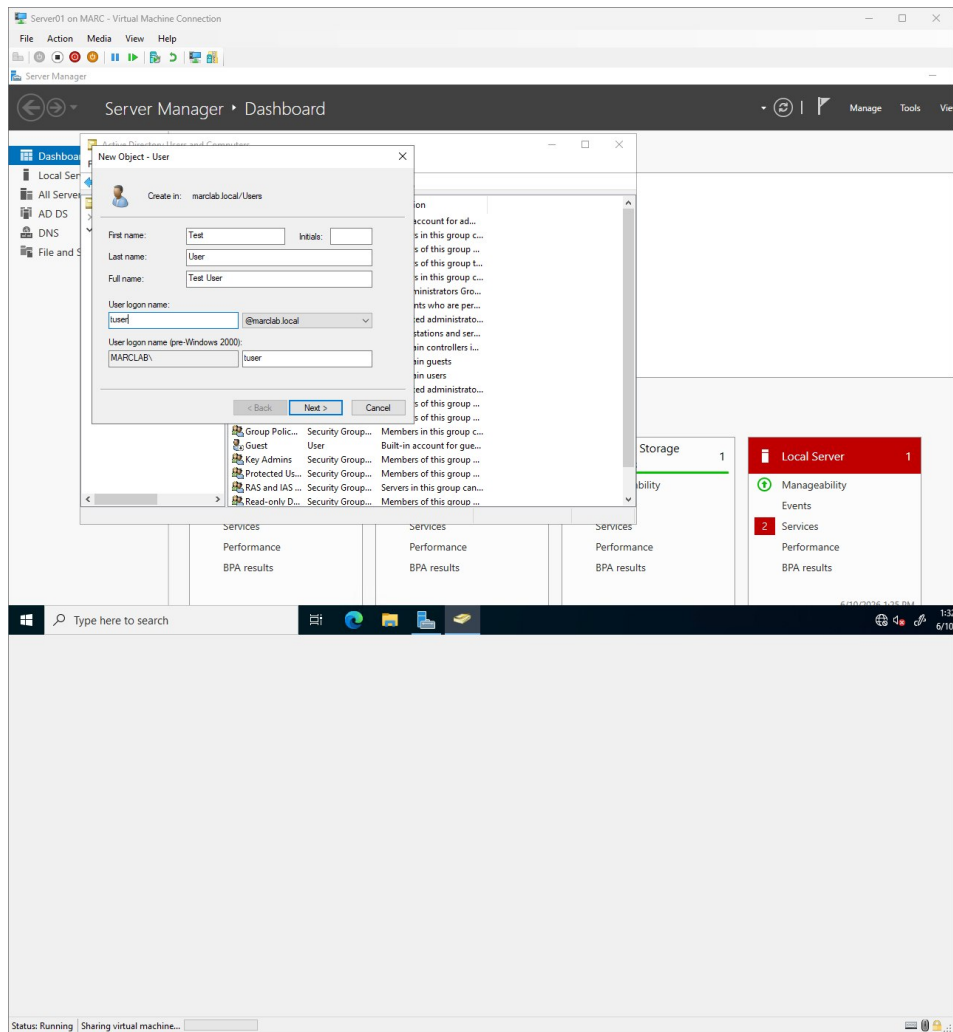
Completed Configuration Tasks

- Created Server01 as the Windows Server domain controller.
- Configured static addressing for the lab domain controller.
- Created the marclab.local Active Directory domain.
- Created Client01 as a Windows 11 client VM.
- Pointed Client01 DNS to Server01.
- Joined Client01 to the marclab.local domain.
- Created OUs for departments, lab users, lab computers, and lab groups.
- Created domain users and security groups for access management practice.

Screenshot Evidence

Duplicate or near-duplicate screenshots were omitted so each screenshot shows a distinct completed task.

1. Creating a Domain User



Created a test domain user in Active Directory Users and Computers.

2. Testing Connectivity and DNS

```
Command Prompt
Pinging 192.168.50.10 with 32 bytes of data:
Reply from 192.168.50.10: bytes=32 time<1ms TTL=128
Reply from 192.168.50.10: bytes=32 time<1ms TTL=128
Reply from 192.168.50.10: bytes=32 time<1ms TTL=128
Reply from 192.168.50.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\localadmin>ping Server01

Pinging Server01.local [fe80::edaa:75c9:c06d:117e%5] with 32 bytes of data:
Reply from fe80::edaa:75c9:c06d:117e%5: time<1ms
Reply from fe80::edaa:75c9:c06d:117e%5: time<1ms
Reply from fe80::edaa:75c9:c06d:117e%5: time<1ms
Reply from fe80::edaa:75c9:c06d:117e%5: time<1ms

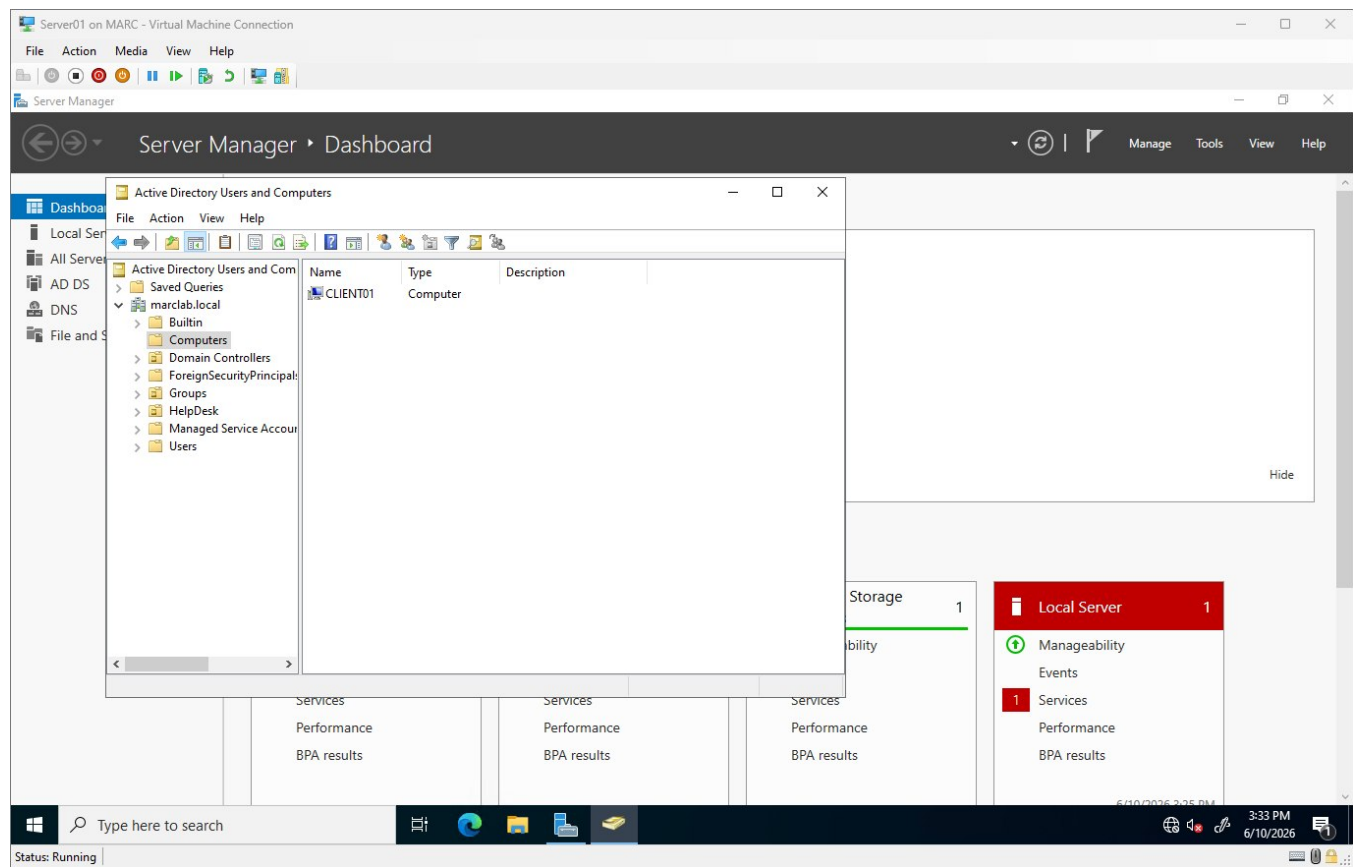
Ping statistics for fe80::edaa:75c9:c06d:117e%5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\localadmin>nslookup marclab.local
DNS request timed out.
    timeout was 2 seconds.
Server:    UnKnown
Address:   192.168.50.10

Name:      marclab.local
Address:   192.168.50.10
```

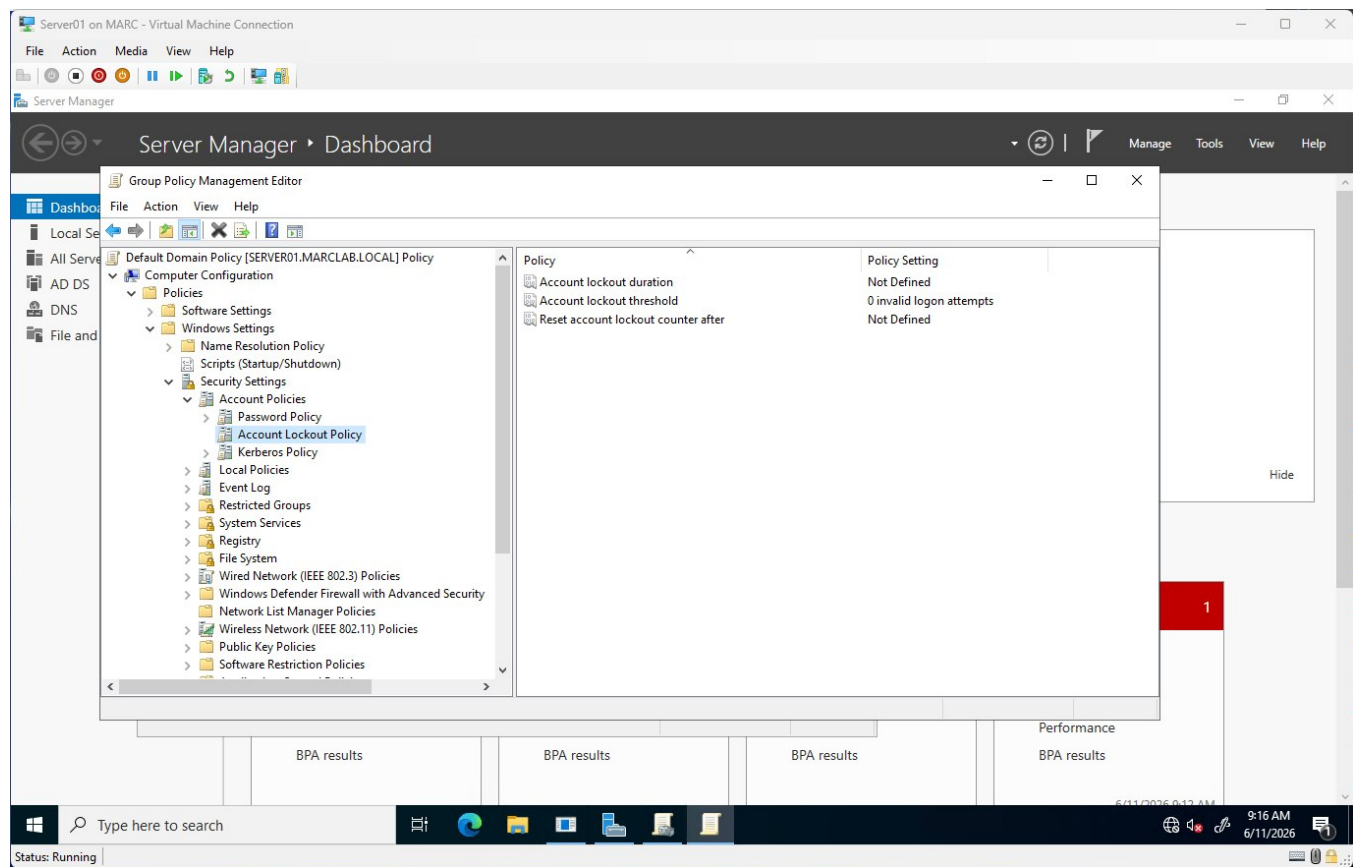
Verified Client01 could reach Server01 by IP address and hostname. nslookup resolved marclab.local to 192.168.50.10.

3. Confirming the Domain Join



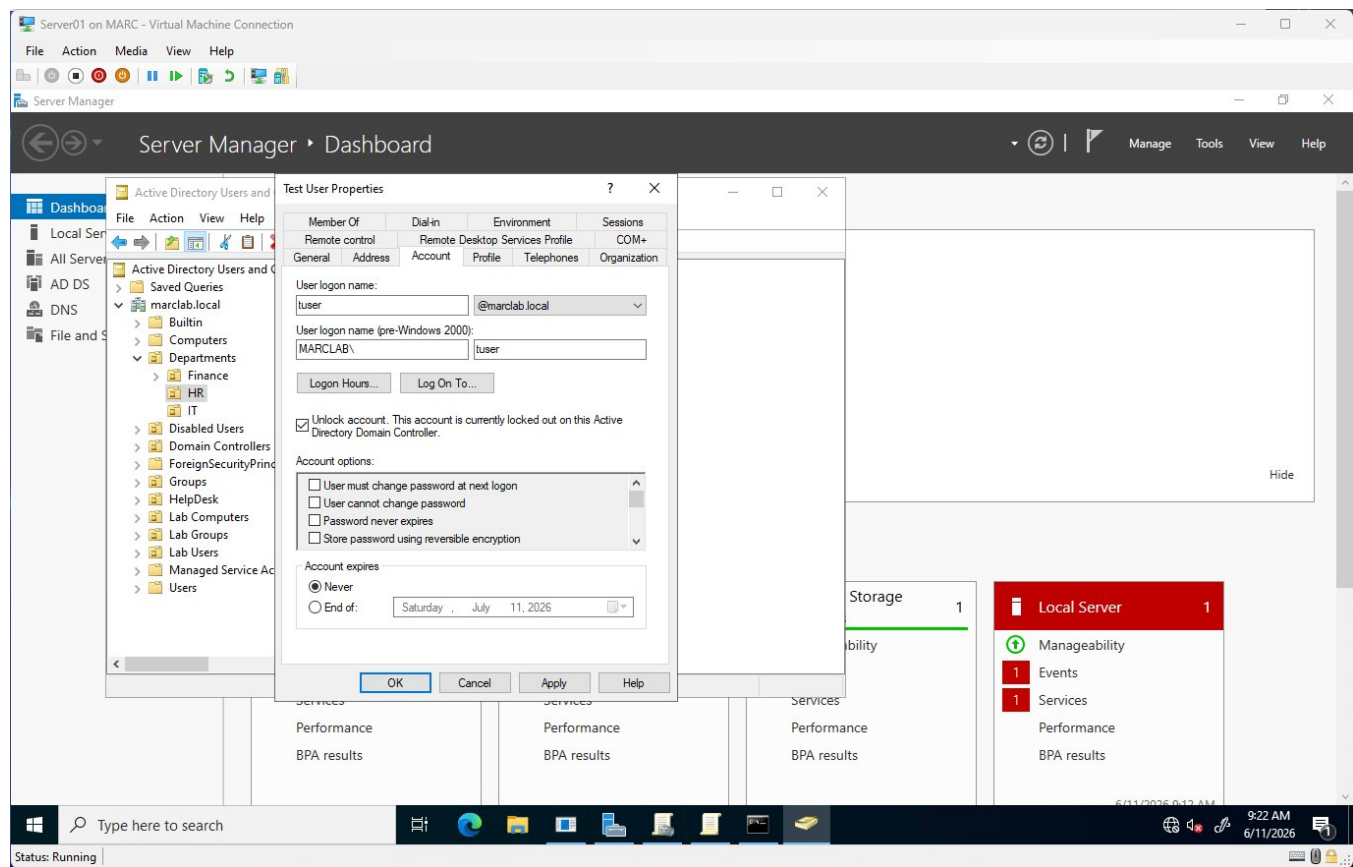
Confirmed the CLIENT01 computer object appeared in Active Directory after joining the marclab.local domain.

4. Reviewing Account Lockout Policy



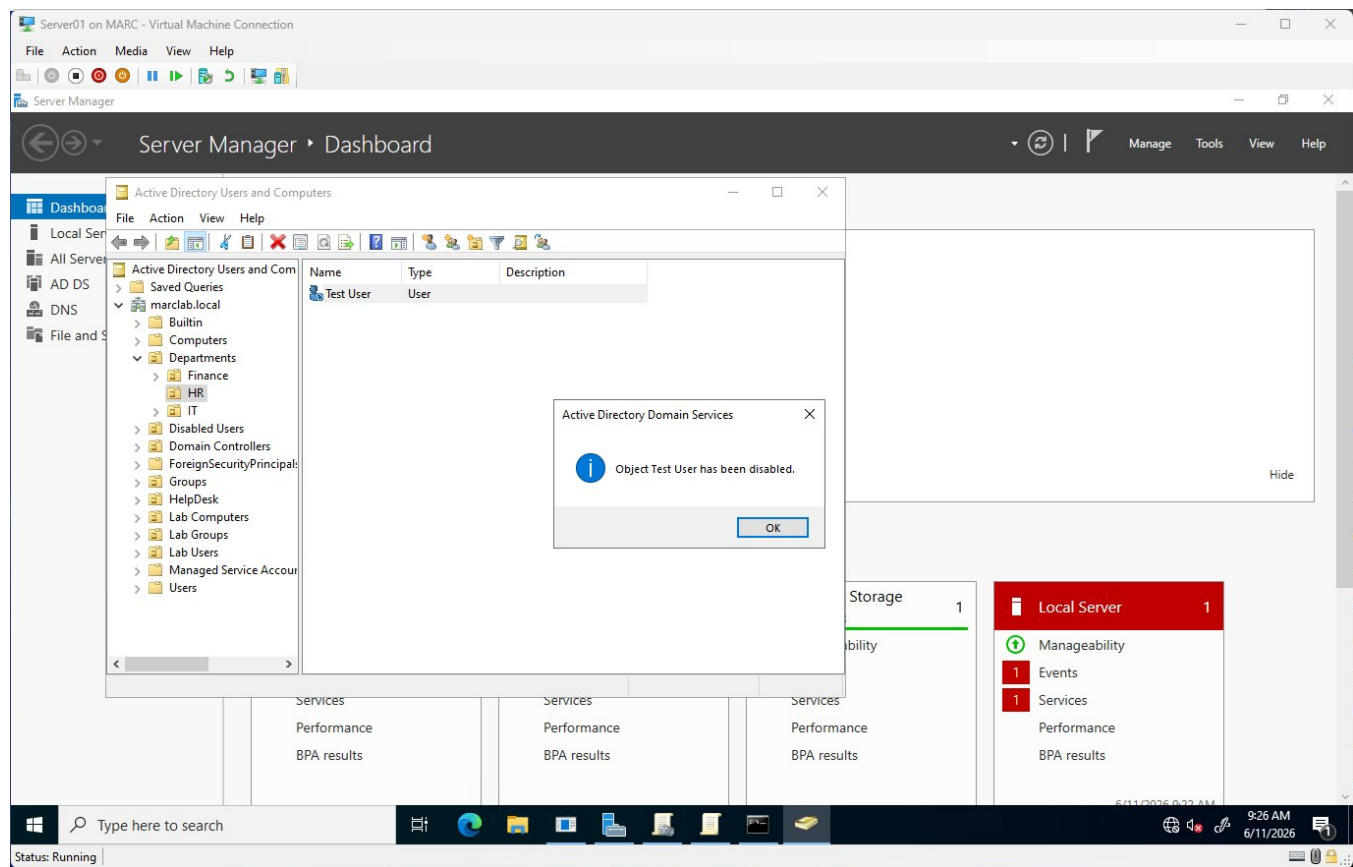
Reviewed account lockout settings in Group Policy Management Editor.

5. Unlocking a User Account



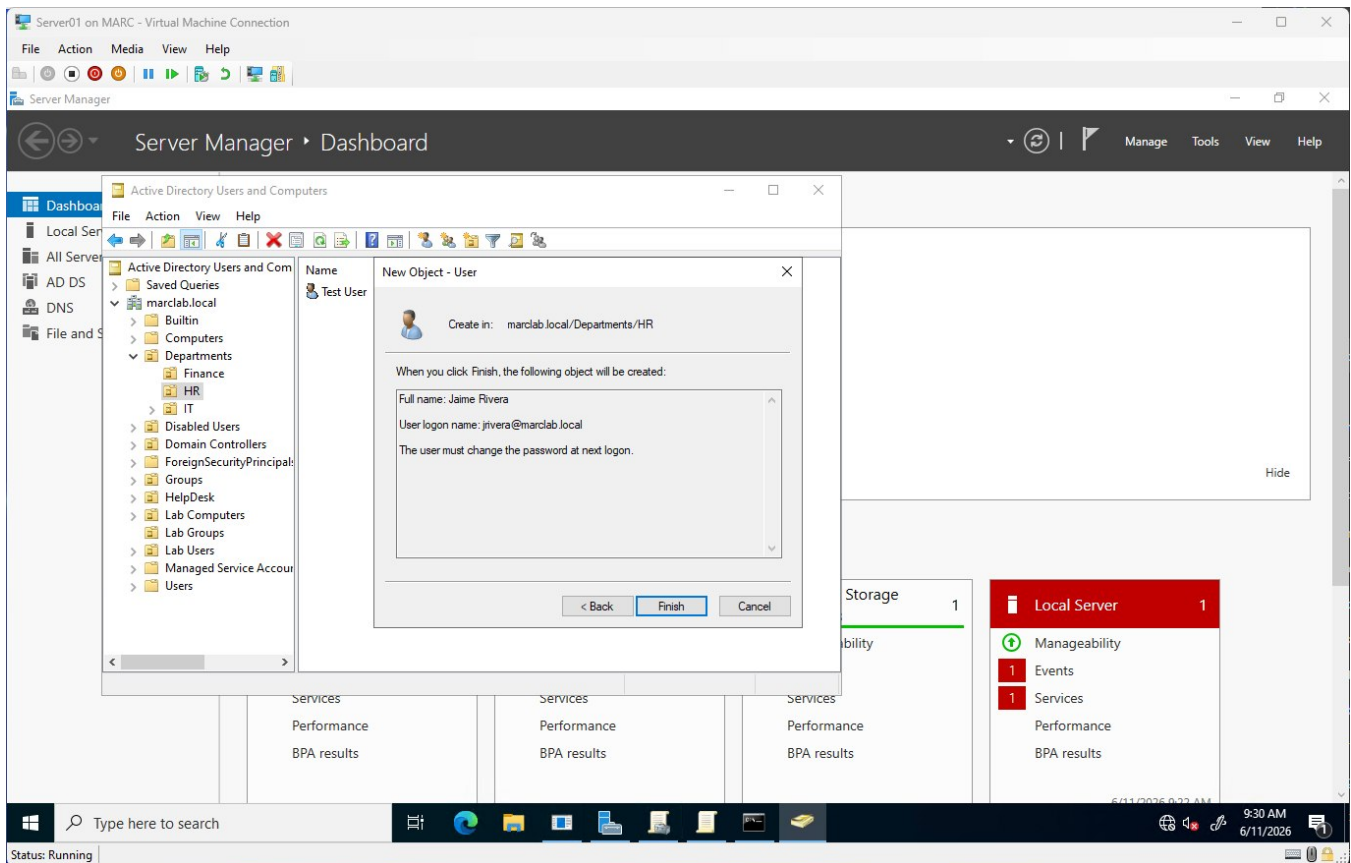
Unlocked the Test User account from the Account tab in Active Directory Users and Computers.

6. Disabling a User Account



Disabled a domain user account to simulate a common access-control/offboarding task.

7. Creating a New Employee Account



Created an HR user account for Jamie Rivera and required a password change at next logon.

8. Verifying Domain Login and Groups

```
Client01 on MARC - Virtual Machine Connection
File Action Media Clipboard View Help
Command Prompt
C:\Users\jrivera>whoami
marclab\jrivera

C:\Users\jrivera>whoami /groups

GROUP INFORMATION
-----

Group Name                                     Type                SID                                     Attri
=====
Everyone                                     Well-known group    S-1-1-0                               Manda
tory group, Enabled by default, Enabled group
BUILTIN\Users                               Alias                S-1-5-32-545                           Manda
tory group, Enabled by default, Enabled group
NT AUTHORITY\INTERACTIVE                     Well-known group    S-1-5-4                               Manda
tory group, Enabled by default, Enabled group
CONSOLE LOGON                               Well-known group    S-1-2-1                               Manda
tory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users             Well-known group    S-1-5-11                              Manda
tory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization               Well-known group    S-1-5-15                              Manda
tory group, Enabled by default, Enabled group
LOCAL                                        Well-known group    S-1-2-0                               Manda
tory group, Enabled by default, Enabled group
MARCLAB\HR_Users                            Group                S-1-5-21-436821564-1239894869-690896841-1108 Manda
tory group, Enabled by default, Enabled group
MARCLAB\Shared_Drive_Users                  Group                S-1-5-21-436821564-1239894869-690896841-1110 Manda
tory group, Enabled by default, Enabled group
Authentication authority asserted identity Well-known group    S-1-18-1                              Manda
tory group, Enabled by default, Enabled group
Mandatory Label\Medium Mandatory Level      Label                S-1-16-8192
```

Used whoami and whoami /groups on Client01 to verify domain login and group membership.

9. Documenting Help Desk Tickets

```
Ticket 001: Password Reset

Issue:
User tuser could not log in and requested a password reset.

Troubleshooting Steps:
Verified the user account in Active Directory Users and Computers.
Reset the password to a temporary password.
Enabled "User must change password at next logon."
Had the user sign in on Client01 and create a new password.

Resolution:
User successfully changed password and logged into the domain.

Status:
Resolved

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Ticket 002: Account Locked

Issue:
User tuser was locked out after multiple failed login attempts.

Troubleshooting Steps:
Opened Active Directory Users and Computers.
Located the user account.
Checked the Account tab and confirmed the account was locked.
Unlocked the account.
Verified the user could log in successfully.

Resolution:
Account was unlocked and user access was restored.

Status:
Resolved

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Ticket 003: New Employee Onboarding

Issue:
New HR employee Jamie Rivera needed a domain account.

Troubleshooting Steps:
Created domain user jrivera in the HR OU.
Assigned temporary password.
Enabled "User must change password at next logon."
Added user to HR_Users and Shared_Drive_Users groups.
Verified successful login from Client01.

Resolution:
New employee account was created and verified.

Status:
Resolved
```

Created written ticket examples for password reset, account lockout, and new user onboarding scenarios.

Mock Help Desk Ticket Examples

Ticket 001 - Password Reset

Issue: User Test User could not log in and requested a password reset.

Troubleshooting/Action Taken: Verified the account in Active Directory Users and Computers. Reset the password to a temporary password and required a password change at next logon.

Resolution: User changed the password and logged into the domain successfully.

Status: Resolved

Ticket 002 - Account Locked

Issue: User Test User was locked out after multiple failed login attempts.

Troubleshooting/Action Taken: Opened the user properties in Active Directory Users and Computers, confirmed the lockout status, and selected the unlock account option.

Resolution: Account was unlocked and the user was able to sign in again.

Status: Resolved

Ticket 003 - New Employee Onboarding

Issue: New HR employee Jamie Rivera needed a domain account.

Troubleshooting/Action Taken: Created the jrivera account in the HR OU, assigned a temporary password, required password change at next logon, and added the user to HR_Users and Shared_Drive_Users.

Resolution: New employee account was created and group membership was verified from Client01.

Status: Resolved

Summary

Built a Windows Server and Active Directory home lab in Hyper-V with a Windows Server domain controller and a Windows 11 domain-joined client. Practiced common help desk tasks including user creation, password resets, account unlocks, account disablement, group membership verification, domain connectivity testing, DNS validation, and help desk ticket documentation.